

# The Spectral-Radius Criterion for All Quadratic Holomorphic Polynomials in Two Free Circular Variables

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**Status.** Complete solution of the finite-dimensional question in Remark 1.5 / Remark 4.7 of Miyagawa, likely valid and pending human review. The result removes every coefficient restriction from the source’s spectral-radius criterion. It is a substantial partial result relative to the broader question asking whether spectrum always equals Brown-measure support.

## 1 Source question and setup

Let  $c_1, c_2$  be freely independent circular operators and

$$P(c_1, c_2) = \sum_{i,j=1}^2 a_{ij} c_i c_j + \sum_{i=1}^2 b_i c_i, \quad A = (a_{ij}) \in M_2(\mathbb{C}), \quad b = (b_1, b_2)^t \in \mathbb{C}^2.$$

For  $\lambda \neq 0$ , set  $A_\lambda = \lambda^{-1}A$  and  $b_\lambda = \lambda^{-1}b$ . Miyagawa’s quadratic-spectrum theorem [1, Theorem 1.4 and Theorem 4.6] defines

$$Q_\lambda = \begin{pmatrix} b_\lambda^* b_\lambda & \text{Tr}(A_\lambda^* A_\lambda) & b_\lambda^* A_\lambda & b_\lambda^t \overline{A_\lambda} \\ 1 & 0 & 0 & 0 \\ \overline{b_\lambda} & 0 & 0 & \overline{A_\lambda} \\ b_\lambda & 0 & A_\lambda & 0 \end{pmatrix}, \quad e_1 = (1, 0, 0, 0, 0)^t, \quad (1)$$

where the block sizes are 1, 1, 2, 2, and proves

$$\lambda \in \text{spec}(P(c_1, c_2)) \iff \lambda = 0 \text{ or } Q_\lambda^n e_1 \not\rightarrow 0. \quad (2)$$

The source observes that the right side of (2) implies  $r(Q_\lambda) \geq 1$ , but leaves open whether the converse always holds. It proves the converse under any of the extra assumptions  $b = 0$ ,  $\overline{A}A$  has no distinct real eigenvalues, or  $A$  is symmetric [1, Remark 1.5 and Remark 4.7].

## 2 Proof intuition

Suppose a dominant eigenvalue of  $Q_\lambda$  were invisible from the orbit of  $e_1$ . A left eigenvector calculation forces it into the lower  $4 \times 4$  block

$$B = \begin{pmatrix} 0 & \overline{A_\lambda} \\ A_\lambda & 0 \end{pmatrix}.$$

The determinant of  $Q_\lambda$  is so large relative to a hypothetical complex or smaller-modulus eigenvalue of  $\overline{A_\lambda}A_\lambda$  that only one case survives: the square of the hidden eigenvalue is the larger of two distinct nonnegative eigenvalues.

where  $e_1 = {}^t(1 \ 0 \ 0 \ 0 \ 0 \ 0)$  and  $Q_\lambda$  is the following  $6 \times 6$  matrix :

$$Q_\lambda = \begin{pmatrix} b_\lambda^* b_\lambda & \text{Tr}(|A_\lambda|^2) & b_\lambda^* A_\lambda & b_\lambda \overline{A_\lambda} \\ 1 & 0 & 0 & 0 \\ \overline{b_\lambda} & 0 & 0 & \overline{A_\lambda} \\ b_\lambda & 0 & A_\lambda & 0 \end{pmatrix}$$

with  $A_\lambda = \lambda^{-1}A$  and  $b_\lambda = \lambda^{-1}b$ .

*Remark 1.5.* In many cases, the condition  $\lim_{n \rightarrow \infty} Q_\lambda^n e_1 \neq 0$  is equivalent to the condition that the spectral radius of  $Q_\lambda$  is greater than or equal to 1. From the Jordan standard form of  $Q_\lambda$ , we can see that if  $\lim_{n \rightarrow \infty} Q_\lambda^n e_1 \neq 0$ , then the spectral radius  $Q_\lambda$  is greater than or equal to 1. Since we take a specific vector  $e_1$ , the converse statement might not be true, although we couldn't find a counterexample. Actually, in the following cases, we can replace  $\lim_{n \rightarrow \infty} Q_\lambda^n e_1 \neq 0$  by the spectral radius  $r(Q_\lambda) \geq 1$ :

- $b = 0$ .
- $\overline{A}A$  doesn't have different real eigenvalues.
- $A$  is symmetric.

When  $P = \frac{1}{2} \sum_{i,j=1}^2 c_i c_j + \frac{1}{2} \sum_{i=1}^2 c_i$  and  $P = c_1 c_2 + c_2 c_1 + c_1 + c_2$  ((A), (B) in Figure 1), the corresponding matrices  $A$  are

$$\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

which are symmetric.

When  $P = c_1^2 + c_2 c_1 - c_2^2 + c_1 + \sqrt{-1} c_2$  ((C) in Figure 1), the corresponding matrix  $A$  is

$$\begin{pmatrix} 1 & 0 \end{pmatrix}$$

Figure 1: The question and the three known coefficient restrictions in Remark 1.5 of the source, PDF page 3.

In that case a unit eigenvector  $x$  is automatically a con-eigenvector:  $A_\lambda x = \overline{\alpha} \overline{x}$ , with  $|\alpha| = r(Q_\lambda)$ . A root of the scalar equation

$$\overline{\alpha} t^2 + (b_\lambda^t x) t = 1$$

has modulus at most  $|\alpha|^{-1/2} \leq 1$ . Thus  $z = tx$  is in the closed unit ball and  $P(z) = \lambda$ . Scalar values of a free polynomial on that ball lie in the spectrum of its left-creation realization by the free Szegő kernel. The source proves that this spectrum equals that of  $P(c_1, c_2)$ , contradicting the assumed invisibility through (2).

### 3 Two lemmas

Write  $L_1, L_2$  for the left creation operators on the full Fock space over  $\mathbb{C}^2$ . For a scalar pair  $z = (z_1, z_2)$ ,  $P(z)$  means the ordinary scalar evaluation of the same noncommutative polynomial.

**Lemma 1** (scalar evaluations are spectral). *If  $\|z\|_2 \leq 1$ , then  $P(z) \in \text{spec}(P(L_1, L_2))$ .*

*Proof.* First suppose  $\|z\|_2 < 1$ . For a word  $w = w_1 \cdots w_n$ , put  $z_w = z_{w_1} \cdots z_{w_n}$  and define the free Szegő vector

$$k_z = \sum_w \overline{z_w} e_w.$$

It is nonzero and belongs to Fock space because  $\|k_z\|^2 = \sum_{n \geq 0} \|z\|_2^{2n} < \infty$ . Moreover  $L_i^* k_z = \overline{z_i} k_z$ , and hence

$$P(L_1, L_2)^* k_z = \overline{P(z)} k_z.$$

Thus  $P(z)$  is in the spectrum. If  $\|z\|_2 = 1$ , apply the strict-ball result to  $rz$  for  $0 < r < 1$  and let  $r \uparrow 1$ . Since spectra are closed and  $P(rz) \rightarrow P(z)$ , the boundary case follows.  $\square$

**Lemma 2** (form of an invisible dominant eigenvalue). *Let  $C \in M_2(\mathbb{C})$  and  $d \in \mathbb{C}^2$ , and let  $Q$  be the matrix in (1) with  $(A_\lambda, b_\lambda)$  replaced by  $(C, d)$ . Suppose  $Q^n e_1 \rightarrow 0$  and  $r(Q) \geq 1$ . Then there are a unit vector  $x \in \mathbb{C}^2$  and  $\alpha \in \mathbb{C}$  such that*

$$\overline{C} \overline{x} = \alpha x, \quad Cx = \overline{\alpha} \overline{x}, \quad |\alpha| = r(Q). \quad (3)$$

*Proof.* Choose an eigenvalue  $\rho$  of  $Q$  with  $|\rho| = r(Q)$  and a nonzero left eigenvector  $v = (v_1, \dots, v_6)$ . Since

$$vQ^n e_1 = \rho^n v_1 \rightarrow 0$$

and  $|\rho| \geq 1$ , one has  $v_1 = 0$ . The second coordinate of  $vQ = \rho v$  then gives  $v_2 = 0$ . Consequently  $(v_3, \dots, v_6)$  is a nonzero left eigenvector, with eigenvalue  $\rho$ , of

$$B = \begin{pmatrix} 0 & \overline{C} \\ C & 0 \end{pmatrix}.$$

Thus  $\rho^2$  is an eigenvalue of  $M = \overline{C}C$ .

Let  $\theta_1, \theta_2$  be the eigenvalues of  $M$ , ordered so that  $|\theta_1| \geq |\theta_2|$ . Their sum is real and their product is  $|\det C|^2 \geq 0$ , so they are either a nonreal conjugate pair or real numbers of the same sign. Expanding  $\det Q$  first along its second row and then along the remaining scalar column gives

$$|\det Q| = T |\det C|^2, \quad T = \text{Tr}(C^* C). \quad (4)$$

As  $|\rho|$  is the largest eigenvalue modulus of the  $6 \times 6$  matrix  $Q$ ,

$$|\det Q| \leq |\rho|^6. \quad (5)$$

If  $\theta_1, \theta_2$  are a nonreal conjugate pair, or if they are equal and nonzero, then  $|\rho|^2 = |\det C|$ . The singular-value inequality  $T \geq 2|\det C|$  makes (4) at least  $2|\det C|^3$ , contradicting (5). The zero repeated-root case would give  $\rho = 0$ . Hence the roots are distinct and real.

If  $\rho^2 = \theta_2$ , then  $\theta_2 \neq 0$  and  $T \geq r(M) = |\theta_1|$ . Equations (4)–(5) give

$$|\theta_1|^2 |\theta_2| \leq T |\theta_1 \theta_2| = |\det Q| \leq |\theta_2|^3,$$

which is impossible because  $|\theta_1| > |\theta_2|$ . Therefore  $\rho^2 = \theta_1$ .

Let  $x$  be a unit eigenvector of  $M$  for the simple real eigenvalue  $\theta_1$ . Conjugating  $Mx = \theta_1 x$  shows that  $\overline{C} \overline{x}$  is another eigenvector for  $\theta_1$ ; it is nonzero because  $\theta_1 = \rho^2 \neq 0$ . Simplicity gives  $\overline{C} \overline{x} = \alpha x$  for some  $\alpha \neq 0$ . Conjugation gives  $Cx = \overline{\alpha} \overline{x}$ , and substitution into  $Mx = \theta_1 x$  gives  $\theta_1 = |\alpha|^2 > 0$ . Finally  $|\alpha| = |\rho| = r(Q)$ , proving (3).  $\square$

## 4 The all-coefficient criterion

**Theorem 1.** *For every  $A \in M_2(\mathbb{C})$ ,  $b \in \mathbb{C}^2$ , and  $\lambda \neq 0$ ,*

$$Q_\lambda^n e_1 \not\rightarrow 0 \iff r(Q_\lambda) \geq 1.$$

*Consequently,*

$$\text{spec}(P(c_1, c_2)) = \{0\} \cup \{\lambda \in \mathbb{C} \setminus \{0\} : r(Q_\lambda) \geq 1\}.$$

*No symmetry, eigenvalue, or vanishing-linear-term hypothesis is required.*

*Proof.* If  $Q_\lambda^n e_1$  does not tend to zero, the Jordan normal form immediately gives  $r(Q_\lambda) \geq 1$ .

Conversely, suppose  $r(Q_\lambda) \geq 1$  but  $Q_\lambda^n e_1 \rightarrow 0$ . Apply Lemma 2 with  $C = A_\lambda$  and  $d = b_\lambda$ , obtaining  $x$  and  $\alpha$ . Since  $x$  is unit,

$$x^t C x = \bar{\alpha}. \quad (6)$$

Put  $\beta = d^t x$ . The two roots of

$$\bar{\alpha} t^2 + \beta t - 1 = 0 \quad (7)$$

have product  $-1/\bar{\alpha}$ . Hence at least one root satisfies

$$|t| \leq |\alpha|^{-1/2} = r(Q_\lambda)^{-1/2} \leq 1.$$

For  $z = tx$ , equations (6) and (7) yield

$$z^t A_\lambda z + b_\lambda^t z = 1, \quad \text{equivalently} \quad P(z) = \lambda.$$

Lemma 1 therefore places  $\lambda$  in  $\text{spec}(P(L_1, L_2))$ . Miyagawa proves  $\text{spec}(P(L_1, L_2)) = \text{spec}(P(c_1, c_2))$  [1, Corollary after Lemma 4.1]. Now (2) implies  $Q_\lambda^n e_1 \not\rightarrow 0$ , a contradiction. This proves the equivalence, and the displayed spectral formula follows from (2).  $\square$

## 5 Verification, scope, and novelty audit

The proof is exact and has no computational dependency. The determinant identity (4) can be checked directly by the two indicated Laplace expansions; the remaining determinant estimates use only  $r(\bar{C}C) \leq \|C\|^2 \leq \text{Tr}(C^*C)$  and  $\text{Tr}(C^*C) \geq 2|\det C|$ . The scalar-evaluation lemma includes the boundary of the ball by radial approximation, so no nonnormal approximate-eigenvector claim is being assumed.

The theorem fully answers the sharper question in Remarks 1.5 and 4.7 and removes all three restrictions listed there. It does not settle the broader question in the introduction asking whether the spectrum of every rational or polynomial function in free semicircular operators equals the support of its Brown measure. In particular, weak convergence of quadratic Ginibre eigenvalue measures identifies the limiting Brown measure but does not by itself exclude spectral regions carrying zero Brown mass; proving support equality would require additional lower-density or Brown-potential control.

A bounded novelty audit searched the run's result, attempt, and proof-gap indexes, arXiv:2603.19528, the exact title and spectral-radius phrasing, and the source's cited quadratic-Ginibre reference. No later theorem resolving the all-coefficient criterion was located. Because the source is very recent and the final external-source refresh was unavailable, novelty is plausible but remains pending specialist review.

Recommended human-review focus is: (1) the block-order and transpose conventions in Lemma 2; (2) the determinant comparison excluding all cases except the positive simple eigenvalue; and (3) the use of the source's spectrum equality and recursion theorem in the last contradiction.

## References

- [1] A. Miyagawa, *The spectra of polynomials in free (semi)circular operators*, arXiv:2603.19528v2 (2026).